

Edgar Salus

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EDUCATION

Cornell University, College of Engineering

B.S. Electrical and Computer Engineering (intended)

Ithaca, NY

Expected May 2030

- **Coursework:** Calculus for Engineers (MATH 1910), Modern Structures (CEE 1160)

Berlin International School

IB Diploma: 41/45 – HL Mathematics: Analysis & Approaches, Physics, German

Berlin, Germany

June 2026

ENGINEERING EXPERIENCE

Stadler Rail

Engineering Intern

Jun 2024 – Jul 2024

Berlin, Germany

- Assembled and wired electrical panels to engineering drawings for three rail-vehicle programs: the **FLIRT H2** hydrogen train, the **RS ZERO**, and the Milan tramway
- Performed underfloor cabling on vehicles in production, working to Stadler's production quality standards

R3 Solutions GmbH

Intern

Aug 2023

Berlin, Germany

- Tested industrial networking hardware and gave feedback on its technical user documentation

PROJECTS & RESEARCH

Hauptstadt Motorsport – STEM Racing (F1 in Schools)

Founder & Team Lead

Nov 2024 – May 2025

Berlin, Germany

- Founded Berlin's first STEM Racing team; led design, manufacture, and test of a CO₂-powered model race car for a 20 m sprint. **Berlin state champion, Best Newcomer, 11th** at the 2025 German National Finals
- Cut run time **19%** (1.410 s → 1.140 s) from regional to national car after a rigid-body model identified mass as the key driver
- Moved production from SLA resin to **SLS PA12 nylon** with an industrial partner, cutting **21.5 g** and offsetting a mandatory 7.1 g halo
- Ran **FEA** on 6 wheel designs (20 N radial load) to trade rotational inertia against deflection; selected a 5-cutout wheel (192 g·mm², 0.014 mm) and full-ceramic bearings after spin tests
- Ran **ANSYS Fluent** CFD with rotating wheels and cross-checked trends in a self-built wind tunnel and on track; raised **EUR 20,000** in sponsorship

Vertical-Axis Wind Turbine Startup Optimization

Independent Researcher – Published, *Int. Journal of High School Research*, Vol. 8, Issue 11

Jun 2025 – Feb 2026

Remote

- Built **6DOF transient CFD** simulations in ANSYS Fluent (realizable $k-\epsilon$, 4.1 m/s inlet, 30% turbulence intensity) so the rotor spun up from rest under aerodynamic torque alone; 3,000+ time steps and 60,000+ iterations per run
- Screened designs with fast steady-state runs using a drag-difference-to-inertia metric before committing to transient runs; varied blade count (2–5), peak offset (0–60 cm), blade depth (1.0–1.4 m), and 3 end-plate iterations
- Final geometry delivered **16% higher peak torque** and 4% more power after 150 s than baseline; traced the shortfall vs. prediction to a 5% increase in rotor inertia

Machine-Learning Classification of Exoplanet Candidates

Independent Researcher – Published, *Curieux Academic Journal*, Issue 56

Jan 2023 – Aug 2024

Remote

- Trained decision-tree, random-forest, and TensorFlow neural-network classifiers on ~10,000 Kepler objects; used XGBoost feature importance to build reduced feature sets
- Showed tree models fell from ~97% to 72–74% accuracy with half the features while the neural network stayed above 95%

LEADERSHIP & HONORS

President, Junior Economic Club Berlin (2025–26): led 20+ events, recruited 30+ members; previously Co-President & CTO

Founder, The Quantum (2023–26): led a 6-person, 4-country team producing physics and astronomy content

Waterloo Cayley Math Contest (2024): top 1.5% worldwide, 1st in Germany; Distinction in Pascal (2023) and Fermat (2025)

Int. Astronomy & Astrophysics Competition (2024): Silver Honour, top 10% of ~5,000 participants

Int. STEM Olympiad (2025): Silver, Technology & Engineering | **German Physics Olympiad** (2023): advanced to Round 2

Junior.ING national engineering competition (2022): 5th place in Germany

TECHNICAL SKILLS

Electrical & Hardware: Panel assembly and wiring to drawings, cable routing, hardware testing, SLS/SLA/FDM 3D printing

Simulation & CAD: ANSYS Fluent (CFD, 6DOF, transient), Autodesk Fusion (CAD, FEA, generative design), SolidWorks

Programming: Python (NumPy, Pandas, Matplotlib, scikit-learn, TensorFlow, XGBoost), L^AT_EX

Languages: German (native), English (native)